

TD4: Logique formelle

Exercice 1:

$$\forall x \exists y (R(x, y) \Rightarrow R(x, f(x)))$$

$$\equiv \forall x \exists y (\neg R(x, y) \vee R(x, f(x)))$$

$$\equiv \forall x (\neg R(x, g(x)) \vee R(x, f(x)))$$

$$\equiv \neg R(x, g(x)) \vee R(x, f(x))$$

$$S \equiv \{ \neg R(x, g(x)) \vee R(x, f(x)) \}$$

2)

$$\forall x (P(x)) \vee (\exists x. (\neg P(x)) \wedge \forall y. Q(y))$$

$$\equiv \forall x (P(x)) \vee (\exists z. (\neg P(z)) \wedge \forall y. Q(y))$$

$$\equiv \forall x, \exists z, \forall y (P(x) \vee (\neg P(z) \wedge Q(y)))$$

$$\equiv \forall x, \forall y (P(x) \vee (\neg P(f(x)) \wedge Q(y)))$$

$$\equiv P(x) \vee (\neg P(f(x)) \wedge Q(y))$$

$$\equiv (P(x) \vee \neg P(f(x))) \wedge (P(x) \vee Q(y))$$

$$S \equiv \{ P(x) \vee \neg P(f(x)), P(x) \vee Q(y) \}$$

$$S \equiv \{ P(x) \vee \neg P(f(x)), P(z) \vee Q(y) \}$$

3)

$$\forall x. (P(x, x) \wedge \exists y (\neg P(x, y) \vee R(y, x)))$$

$$\equiv \forall x, \exists y. (P(x, x) \wedge (\neg P(x, y) \vee R(y, x)))$$

$$\equiv \forall x. (P(x, x) \wedge (\neg P(x, f(x)) \vee R(f(x), x)))$$

$$\equiv P(x, x) \wedge (\neg P(x, f(x)) \vee R(f(x), x))$$

$$S \equiv \{ P(x, x), \neg P(x, f(x)) \vee R(f(x), x) \}$$

$$S \equiv \{ P(x, x), \neg P(e, f(e)) \vee R(f(e), e) \}$$

4)

$$\exists x, \exists y, \exists z. (R(x, y) \wedge R(y, z) \wedge R(z, x))$$

$$\equiv R(a, b) \wedge R(b, c) \wedge R(c, a)$$

$$S \equiv \{R(a, b), R(b, c), R(c, a)\}$$

Exercice 2:

1)

$$S \equiv \{Q(\underline{f(a)}, g(x)), Q(\underline{y}, y)\} \quad \sigma = \emptyset$$

$$D_1 = \{f(a), y\} \quad \sigma = \{f(a)/y\}$$

$$S \equiv \{Q(\underline{f(a)}, g(x)), Q(\underline{f(a)}, \underline{f(a)})\}$$

$$D_2 = \{g(x), f(a)\} \quad \text{Arrêt avec échec car } D_2 \text{ ne contient pas de variables}$$

→ non unifiable

2)

$$S \equiv \{G(\underline{x}, f(b, x)), G(\underline{y}, f(z, g(x)))\} \quad \sigma = \emptyset$$

$$D_1 = \{x, y\} \quad \sigma = \{x/y\}$$

$$S \equiv \{G(\underline{x}, f(\underline{b}, x)), G(\underline{x}, f(\underline{z}, g(x)))\}$$

$$D_2 = \{b, z\} \quad \sigma = \{x/y, b/z\}$$

$$S \equiv \{G(\underline{x}, f(\underline{b}, \underline{x})), G(\underline{x}, f(\underline{b}, g(x)))\}$$

$$D_3 = \{x, g(x)\} \quad \text{Arrêt avec échec car la variable } x \text{ figure dans le terme } g(x)$$

→ non unifiable

3)

$$S \equiv \{P(\underline{f(a)}, x), P(\underline{x}, a)\} \quad \sigma = \emptyset$$

$$D_1 = \{f(a), x\} \quad \sigma = \{f(a)/x\}$$

$$S \equiv \{P(\underline{f(a)}, \underline{f(a)}), P(\underline{f(a)}, a)\}$$

$$D_2 = \{f(a), a\} \quad \text{Arrêt avec échec car } D_2 \text{ ne contient pas de variable}$$

→ non unifiable

4)

$$S = \{P(\underline{x}, z, y), P(\underline{w}, u, \underline{w}), P(a, u, u)\} \quad \sigma = \emptyset$$

$$D_1 = \{x, w, a\} \quad \sigma = \{a/x, a/w\}$$

$$S = \{P(a, z, y), P(a, u, a), P(a, u, u)\}$$

$$D_2 = \{z, u, u\} \quad \sigma = \{a/x, a/w, u/z\}$$

$$S = \{P(a, u, y), P(a, u, a), P(a, u, u)\}$$

$$D_3 = \{y, a, u\} \quad \sigma = \{a/x, a/w, u/z, a/y, a/u\}$$

$$S = \{P(a, a, a), P(a, a, a), P(a, a, a)\}$$

→ Unifiable

Exercice 3:

$$C_1 = G(a, x, b) \vee R(x, y)$$

$$C_2 = \neg G(T, f(T), y) \vee P(z, T)$$

$$\text{Soit } S = \{G(a, x, b), G(T, f(T), y)\} \quad \sigma = \emptyset$$

$$D_1 = \{a, T\} \quad \sigma \leftarrow \sigma_0 \quad \{a/T\}$$

$$S = \{G(a, x, b), G(a, f(a), y)\}$$

$$D_2 = \{x, f(a)\} \quad \sigma \leftarrow \sigma_0 \quad \{f(a)/x\}$$

$$S = \{G(a, f(a), b), G(a, f(a), y)\}$$

$$D_3 = \{b, y\} \quad \sigma \leftarrow \sigma_0 \quad \{b/y\}$$

$$S = \{G(a, f(a), b), G(a, f(a), b)\}$$

→ S est unifiable avec $\sigma = \{a/T, f(a)/x, b/y\}$

$$C_{1\sigma} = G(a, f(a), b) \vee R(f(a), b)$$

$$C_{2\sigma} = \neg G(a, f(a), b) \vee P(z, a)$$

$C = R(f(a), b) \vee P(z, a)$ est le résolvant de C_1 et C_2